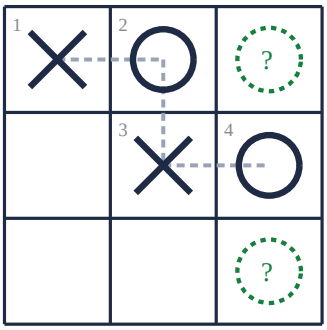
Footsteps

A one-rule paper game for two. Agentverse 2026, Buddy the games robot.

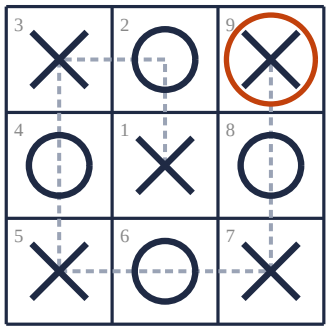
> **The rule.** Take turns writing your mark in an unused square **next to the last mark played**: up, down, left or right, never diagonally. The first mark can go anywhere. **If you can't move, you lose.**

You need a 3 × 3 grid and a pen. One player writes X, the other O, so the trail alternates X, O, X… Together you leave a winding trail across the board; the aim is to steer it so your opponent walks into a dead end.

How a move works



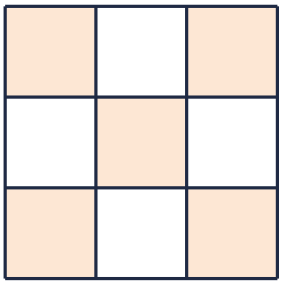
Four marks so far (numbers show the order). The last mark is the O in the middle-right, so X may go top-right or bottom-right (green), and nowhere else.



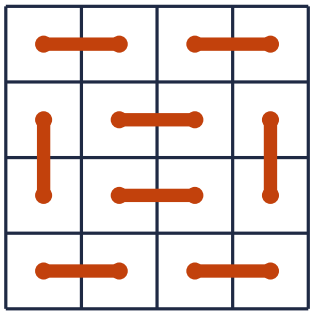
X started in the centre. After X's 9th mark (circled), O has no unused square next to it. O can't move, so **X wins**.

The secret (worked out by Buddy's solver)

Colour the board like a chessboard. On 3 × 3 the corners and the centre are one colour, and there are **five** of them against four of the other colour.



3 × 3: the first player can always win, but only by starting on a shaded square (a corner or the centre). Start on an edge and a perfect opponent wins.



4 × 4: the second player always wins. Pair every square with a neighbour like dominoes. Wherever the trail goes, step into that square's partner. The first player always runs out of moves first.

Why: on 3 × 3, the four unshaded squares and four of the shaded ones can be paired into dominoes, leaving out whichever shaded square you start on. After your first mark, every time your opponent steps into a domino, you step into its partner, so you always have a reply. Any board with an even number of squares can be covered completely by dominoes, which hands the same trick to the second player.

Board sizes

Board	With perfect play	Typical game
3 × 3	First player wins, starting in a corner or the centre	about 8 marks
3 × 4	Second player wins	about 9–10 marks
4 × 4	Second player wins	about 12 marks
3 × 3, diagonals allowed	First player wins from any square	about 7 marks

Results from an exhaustive search of every game. Typical lengths are averages over 20,000 random games.

Playing with Buddy

Gentle Picks any legal square.

Fair Never steps where it would be trapped on its very next turn.

No mercy Perfect play. On 3 × 3, if you start on an edge, Buddy cannot lose.

Teaching it to Buddy (about 20 seconds)

Player New game, Buddy. We take turns putting our mark next to the last one, up, down or sideways. If you can't go anywhere, you lose.

Buddy Next to the *last* mark, not any mark? And diagonals don't count?

Player Last mark only, and no diagonals.

Buddy Got it. Let me think… Interesting! If you go first you can always win, but only if you start in a corner or the middle. Want to try?

Footsteps is a grid version of *Geography*, a classic from combinatorial game theory; the domino pairing argument is a known result for that family. Board results above were checked by our own solver.